

Predator/Prey Simulation Coursework Report

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1. Simulation Description

1.1 Species included

- Predators: Wolf, Bear, Coyote
- Prey: Sheep, Deer, Squirrel
- Plant resource (shown in green)

1.2 Behaviour and interactions

Predators search adjacent cells and eat the first living prey found. All predator species hunt the same prey types, so they directly compete for food resources in the same environment. All prey species feed on plants and have explicit food values which is then used by predators these food values are:

Sheep = 10, Deer = 20, Squirrel = 5. If prey can't eat the food on a given step due to the randomiser not working, they can still move to another free adjacent location.

Plants are placed in any non-animal starting cell, plant cells are treated as free/replaceable locations, and plants are restored when animals move away or die. This maintains a persistent plant baseline in the field. However, having said this, it is not guaranteed that a Prey will eat a plant all the time due to this logic. I have implemented a random chance that the prey will eat the plant. This allows the prey to die due to starvation as well.

1.3 Disease model

Each species stores disease state with a Boolean sick variable and a fixed sickness duration. Animals become sick using a gene-controlled probability, and infection can spread only to neighbouring animals of the same species.

2. Implemented Extension Tasks

2.1 General Requirements

1. Six acting animal species are implemented, including at least two predators and at least two prey.
2. Plants are simulated as green cells, initialized in all non-animal starting cells, treated as free cells, and restored when animals die.

3. Predator competition is implemented: Wolf, Bear, and Coyote all consume prey.
 4. All prey eat plants and expose food values through the Prey interface.
 5. Disease is implemented with Boolean state, fixed duration, and same-species spread mechanics.
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2.2 Natural Selection and Genetics

1. Each animal has a gene that encodes breeding age, life span, breeding probability, litter size, disease probability, and metabolism.
2. Gene outputs are constrained to the limits after decoding and mutation.
3. Only the initial population receives randomly generated genes.
4. Animals are assigned male/female with approximately 50:50 probability.
5. Mating requires neighbouring male/female animals of the same species.
6. Offspring are generated with crossover (first half from one parent, second half from the other).
7. Mutation is applied per gene position with 20% chance; selected positions mutate by +1 if current position is not 9 or -1 if current position is not 0 before fixing to attribute limits.
8. Metabolism is gene-driven and subtracted from food level each step instead of using a fixed value.

2.3 Additional implementation notes

- The simulation includes species specific disease spread chances and sickness durations.

3. Summary

The completed simulation satisfies the coursework requirements by combining a multi-species food chain, plant dynamics, disease transmission, and evolutionary genetics. Interacting pressures from predation, hunger, disease, and inheritance produce changing population outcomes across generations.